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Polyethylene glycol/graphene oxide aerogel shape-stabilized phase change materials for photo-to-thermal energy conversion and storage via tuning the oxidation degree of graphene oxide

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ABSTRACT

Polyethylene glycol (PEG)/graphene oxide aerogel (GA) composite phase change materials (PCMs) were prepared by introducing PEG into GAs from graphene oxide (GO) with different oxidation degree via vacuum impregnation. The structures of GAs were tuned by the oxidation levels of GO. A series of characterizations were used to analyze the chemical structure of GOs, including X-ray diffraction (XRD), Fourier transform infrared spectrum (FTIR), X-ray photoelectron spectroscopy (XPS) and Raman spectroscopy analysis. Structural analyses confirmed that the oxygenated functional groups increased and the hydroxyl groups were transformed into carboxyl and epoxy groups with increasing oxidation level. In addition, the graphitic nature of GO decreased while the sp^3 domains of GOs increased owing to the disruption of the graphitic stacking order. Morphology analysis showed that the breakage of graphene sheet became more serious with the oxidation level increasing. When GAs prepared with GOs of higher oxidation levels were used, the composite PCMs showed excellent shape-stability during phase change and excellent thermal repeatability. The change of dimension for PGA6-40 heated from 35 °C to 150 °C was negligible under the load of a constant force (7 N). Efficient photo-to-thermal energy conversion and storage was realized in the composite PCMs.

1. Introduction

The sustainable development in energy is of critical importance for human beings because a great amount of energy available is limited, intermittent and unbalanced in nature. The increase of greenhouse gases and the shortage of fossil fuels also drive the development and efficient utilization of various kinds of renewable energy. Therefore, energy storage plays a crucial role in saving energy and alleviating the contradiction between the demand and the supply of energy. Thermal energy storage (TES), one popular method of energy storage, can be achieved by sensible heat thermal energy storage, latent heat thermal energy storage and the chemical energy storage of the reversible chemical reaction. Among them, the applications of TES are based on the latent heat thermal energy storage (LHTES) of phase change materials (PCMs) at present, and PCMs have become an integral part of the LHTES system with high energy storage density and isothermal characteristics during heat storing and releasing [1–5].

During melting or solidification, PCMs can store or release a large amount of energy at a nearly constant temperature [6]. As advanced energy-saving materials with so many advantages, PCMs have exhibited important application value and broad development prospects in many fields, including thermal energy management system [3,5,7], aerospace, solar energy utilization [8–11], textile industry [12,13], thermal storage building [14–16], etc. [17,18].

There are many substances that can be used as PCMs, which can be divided into four categories according to the phase change process: solid–solid PCMs, solid–liquid PCMs, solid–gas PCMs and liquid–gas PCMs. Nowadays, solid–liquid PCMs are more widely studied and applied. Solid-liquid PCMs exhibit a solid–liquid transition during phase change, and the liquid phases show a certain fluidity which restricts their applications. Shape stabilization of the PCMs has been proved to be a solution. Shape-stabilized PCMs comprised of working medium and supporting materials have been widely explored, in which the supporting structures and materials can prevent the working medium from leaking during phase transition since 1990s [19]. Many microencapsulation methods of PCMs have been developed from polymer shells to graphite based materials (expanded graphite (EG) [11,20], graphene oxide (GO)

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Nomenclature

PEG	polyethylene glycol
PCM	phase change material
XRD	X-ray diffraction
XPS	X-ray photoelectron spectroscopy
LHTES	latent heat thermal energy storage
AC	active carbon
SEM	scanning electron microscopy
GA	graphene oxide aerogel
GO	graphene oxide
FTIR	Fourier transform infrared
TES	thermal energy storage
EG	expanded graphite
T_m	melting temperature
3D	three-dimensional
x	the mass ratio of $KMnO_4$ and EG during the preparation of GOs
y	the concentration of GO solutions used to prepare GAs
GOx	the obtained GOs with different degree of oxidation
GAx-y	the obtained GAs with different degree of oxidation and different concentration
PGAx-y	the obtained PEG/GA composite PCMs
T_{co}	onset crystallization temperature
T_{ce}	end crystallization temperature
T_{cp}	peak crystallization temperature
ΔH_c	crystallization enthalpy
T_{mo}	onset melting temperature
T_{me}	end melting temperature
T_{mp}	peak melting temperature
ΔH_m	melting enthalpy

[21,22] and graphene [23,24]) and foam structure (diatomite, active carbon (AC), graphene aerogel and carbon aerogel) [25–28]. For instance, Mehrali et al. [21] prepared a form-stable composite PCMs by vacuum impregnating paraffin into graphene oxide sheets and the composite PCMs contained 48.3 wt.% of paraffin without leakage of melted PCM. Wang et al. [29] fabricated three types of polyethylene glycol (PEG) based shape-stabilized PCMs with various pore structures of EG, AC and ordered mesoporous carbon (CMK-5). The highest content of PEG without leakage was about 70 wt.% for AC, 90 wt.% for both EG and CMK-5, respectively. A shape-stabilized PEG/GO composite PCMs obtained by Qi et al. [22] exhibited an extremely high mass percentage of PEG as high as 96 wt.% with no leakage up to the temperature as high as 150 °C which was far higher than the melting temperature of PEG. Owing to the high weight percentage of PEG in the composite PCMs, the energy storage density of shape-stabilized composite PCMs based on PEG was guaranteed.

PEG is one of the most promising solid–liquid organic PCMs due to its relatively high latent heat of fusion, isothermal phase transition behavior, suitable phase transition temperature range, non-corrosiveness, non-toxicity, and no decomposition in its melting/freezing temperature range. However, the leakage of liquid phase above the melting temperature (T_m) is the major drawback of PEG, which must be microencapsulated in some specially sealed containers [30,31]. Karaman et al. [32] prepared form-stable PEG/-diatomite composite PCMs by incorporating PEG in the pores of diatomite and found composite PCM with 50 wt.% of PEG exhibited the excellent shape-stability without the leakage of melted PEG. Zhao et al. [33] fabricated epoxy/PEG shape-stabilized PCMs by in situ reactive blending and the maximum mass fraction of PEG dispersed into epoxy matrix to form epoxy/PEG shape-stabilized PCMs was identified as 70 wt.%.

Recently, the use of three-dimensional (3D) materials as supporting materials has been paid more attention to, especially, the carbonous aerogels [26,34–39]. Graphene oxide aerogel (GA) with a 3D framework formed by the self-assembly of GO sheets shows ultra-low density and high porosity, which is advantageous for the impregnation of PEG. Molten PEG is introduced into the pore space of 3D framework by the capillary force. Meanwhile, the oxygen containing functional groups on GAs can form hydrogen bonds with PEG, further preventing the leakage [40]. There are a great number of oxygenated functional groups on GA, which can be tuned by the oxidation level of GO. It is for sure that the variation of oxidation level of GO would show great influence on the formation and structure of the resulting GAs. However, the relationship between the oxidation degree of GO and the structure of the corresponding GAs has not caught enough attention. As far as we know, the effect of the oxidation degree of GO on the performances of composite PCMs has rarely been investigated, especially the shape-stability of organic PCMs.

Therefore, in the present study, different GOs were synthesized by adjusting the dosages of oxidant. Then, PEG/GA composite PCMs were prepared, and the shape-stability and phase change behaviors of pure PEG and PEG/GA composite PCMs with different concentration of GAs from GO with different degrees of oxidation were characterized, aiming at revealing the relationship among oxidation degree of GOs, the structure of GAs and properties of composite PCMs. It is found that all the composite PCMs show outstanding thermo-physical properties and thermal reliability. When the degree of oxidation increases, the strength of the hydrogen bonds is higher and the movement of the molten PEG chain will be more confined, leading to better shape-stability for PEG/GA composite PCMs during phase change.

The inexhaustible solar energy is the cheapest and greenest saving energy. Therefore, the conversion, storage and application of solar energy have been widely concerned [41,42]. Various carbon and carbon-based materials have been studied for driving solar energy conversion and storage due to their unique structure and properties on sunlight absorption and conversion [43–46]. In this study, it is found that GAs endow the composite PCMs with good light absorption and conversion ability. Compared with pure PEG, the composite PCMs can also realize effective photo-to-thermal energy conversion and storage.

2. Experimental section

2.1. Materials

Expandable graphite (Laixi Nanshu Fada Graphite Company, Qingdao, China) with an average particle size of 200 meshes and a purity of over 99.9% was used in the form of expanded graphite after thermal expansion. The number-average molecular weight of the working substance, PEG (Aladdin Reagent (Shanghai, China)), was 10,000 g/mol. Concentrated sulfuric acid (H_2SO_4), Phosphoric acid (H_3PO_4), potassium permanganate ($KMnO_4$), hydrogen peroxide (H_2O_2) and hydrochloric acid (HCl) were purchased from Chengdu Kelong Chemical Reagent Factory (Chengdu, China). All the chemicals were of analytical reagent grade and used without further purification. Distilled water was used throughout this work.

2.2. Preparation of GO

GOs with different oxidation degrees were prepared from EG powder with an improved Hummers' method [47]. 6.0 g EG was added to a mixture of 9:1 (720 mL/80 mL) concentrated H_2SO_4/H_3PO_4. Subsequently, $KMnO_4$ was slowly added to the mixture under stirring at 35–40 °C. Then the mixture was stirred constantly at 50 °C for 12 h. The resulted mixture after cooling down to room temperature was poured into ∼1000 mL of ice water. Approximately 30 mL of 30% H_2O_2 was dropped into the reaction system under continuous stirring until no air bubbles appeared. The mixture was then centrifuged to remove acids. The precipitate mixture was washed and centrifuged with 5% solution of HCl for three times to remove residual metal oxides. Next the mixture was washed and centrifuged with deionized water until the pH of decantate was close to 7. Deep brown substances were obtained by centrifugation and vacuum freeze-drying. The degree of oxidation of GO was tuned by changing the amount of $KMnO_4$ from 3 g, 6 g, 12 g, 18 g to 36 g. The mass ratios of $KMnO_4$: EG were 0.5, 1, 2, 3, 6, and the obtained GOs with different degrees of oxidation were labeled as GO0.5, GO1, GO2, GO3 and GO6, respectively.

2.3. Preparation of GO aerogels

5 mg/mL, 10 mg/mL, 15 mg/mL, 20 mg/mL and 40 mg/mL of GO solutions were prepared under ultrasonication treatment for 1 h. The solutions with GOs of different degrees of oxidation and different concentrations were poured into a cylindrical mold followed by freeze-drying to prepare GAs. The obtained GAs were marked as GA0.5-5, GA0.5-10…, GA3-15, GA3-20…, GA6-40 (GAx-y), respectively, in which x and y represented the mass ratio of $KMnO_4$ and EG during the preparation of GOs and the concentration of GO solutions used to prepare GAs, respectively.

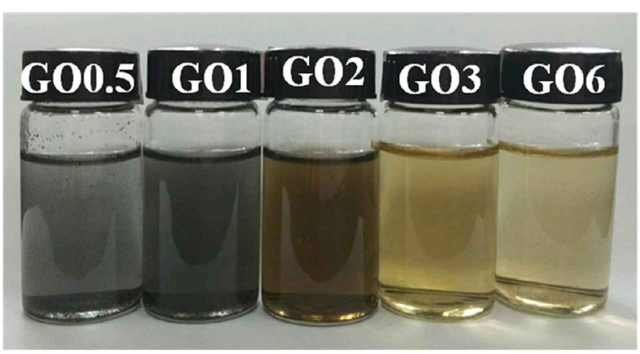

Fig. 1. Photographic images of GO samples (0.2 mg/mL) with different degrees of oxidation.

2.4. Preparation of composite PCMs

The composite PCMs were obtained by a vacuum impregnation method. First, PEG was poured into culture dishes. The culture dishes were then placed in a 90 °C water bath. After PEG powder melted completely, the culture dishes with PEG melt were placed in a 90 °C vacuum drying oven. Vacuum pumping was stopped until no bubbles were created in the PEG melt. Then the prepared GAs were placed into PEG melt for 3 hours with vacuuming, and GAs were absolutely impregnated with PEG melt. At last, the samples were cooled at room temperature and polished by sandpaper to remove extra PEG on the sample surface. The obtained PEG/GA composite PCMs were labeled as PGAx-y.

2.5. Characterizations

X-ray diffraction (XRD) patterns of GOs with different oxidation degrees were performed on a DX-1000 diffractometer using Cu K α radiation (λ = 0.154 nm) at a 40 mA current and a 40 kV voltage. Samples were scanned at a scanning speed of 10° min^{-1} in the range of diffraction angle 2θ = 5–50°. Fourier transform infrared (FTIR) spectroscopy of the GOs were recorded on a Nicolet 6700 FTIR spectrometer (Nicolet Instrument Company, USA) in the wavenumber range of 4000–400 cm^{-1}. X-ray photoelectron spectroscopy (XPS) measurements were carried out on an XSAM800 (Kratos Company, UK) instrument with Al K α radiation (hυ = 1486.6 eV) and casaXPS software was used to calculate the atomic concentrations and separate the overlapped peaks. A DXRxi Raman Microscope (Thermal Scientific, USA) was used to obtain Raman spectra at an excitation wavelength of 532 nm. The morphologies of the GAs were characterized via a field-emission scanning electron microscopy (SEM) instrument (JEOL JSM-5900LV, Japan) at a 20 kV accelerating voltage.

To characterize the shape-stabilization of PCMs with GAs from GOs of different degrees of oxidation, samples were placed on a hot plate, following by taking digital photos of the samples at different temperatures. Dynamic rheometer (AR2000EX, TA instruments, USA) was also utilized to evaluate the shape stability of the samples. Temperature ramp and constant normal force were set to be 2 °C min^{-1} from 35 °C to 150 °C and 7 N. Samples were enclosed in an aluminum pan to measure their thermal properties by differential scanning calorimeter (DSC) using nitrogen as the purge gas in a TA Q20 instrument (USA) at a heating rate of 10 °C min^{-1}. A CELNP2000 optical power meter (Ceaulight, China) and CEL-HXUV300 xenon lamp were used to carry out the photo-to-thermal energy conversion test. Simultaneously, a paperless recorder with thermocouples (OMEGA) was used to record the temperature of samples.

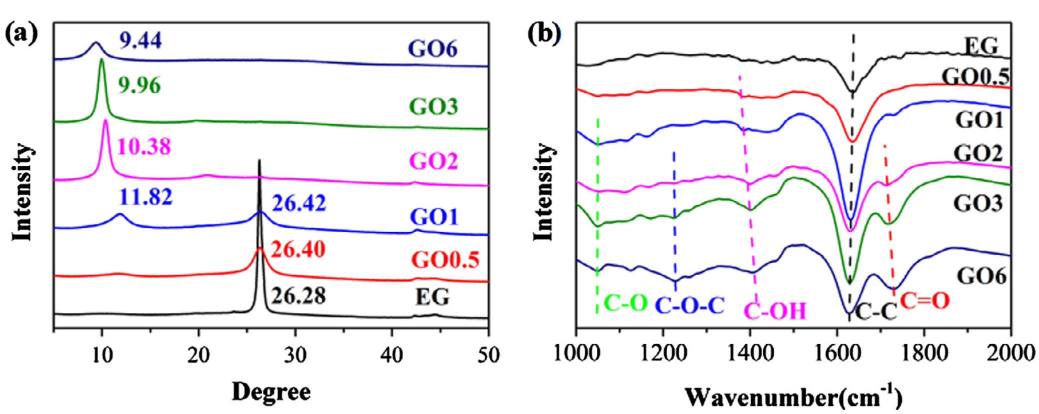

Fig. 2. XRD and FTIR patterns of EG and GOs with different degrees of oxidation.

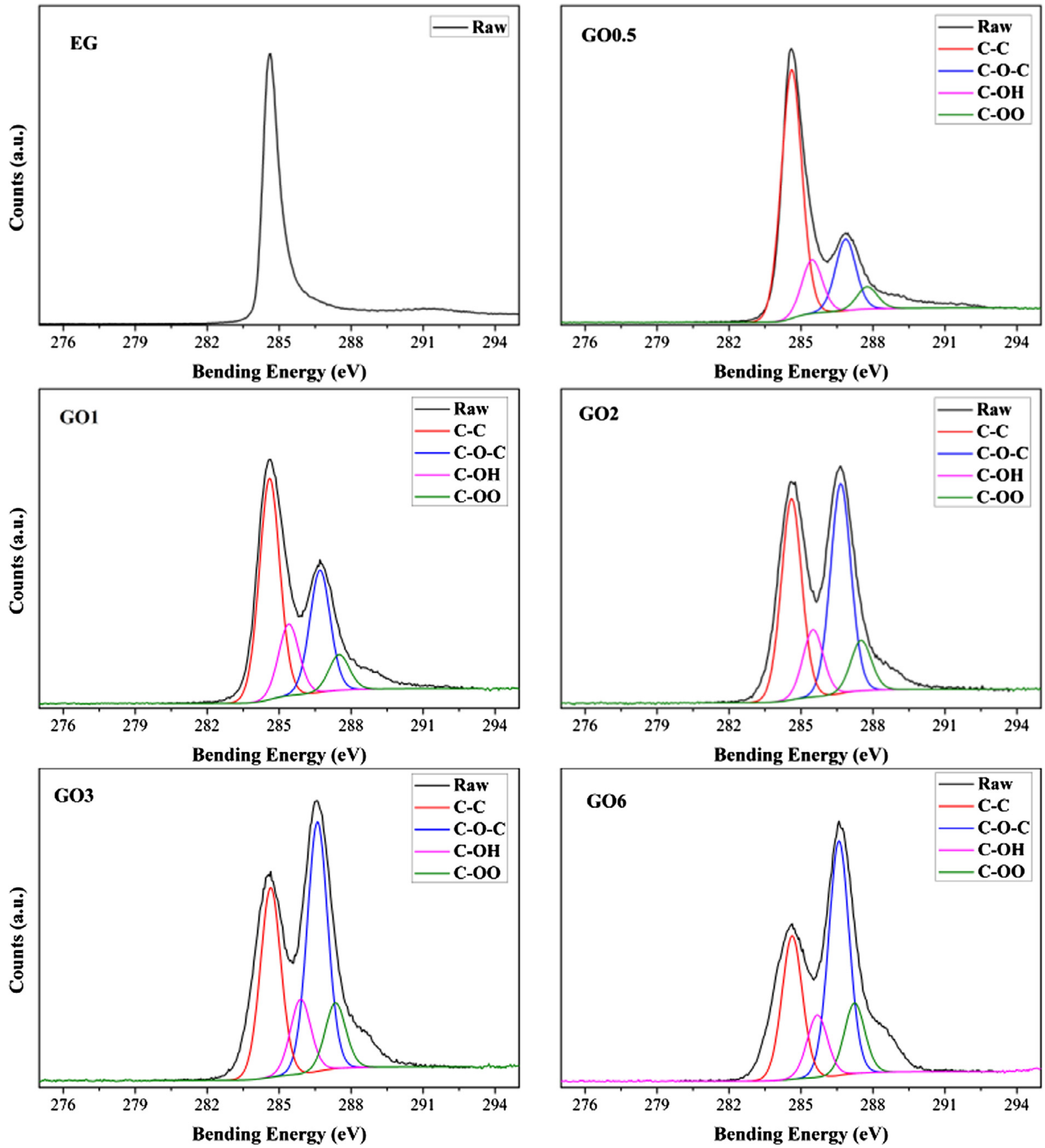

Fig. 3. XPS spectra of GO with different degrees of oxidation.

3. Results and discussion

3.1. The chemical and structural analysis of GOs

In this study, GOs with 5 different oxidation levels are synthesized with a modified Hummers' method by adjusting the dosage of oxidants. The photographic images of 0.2 mg/mL of all these GO suspensions are displayed in Fig. 1. We can see the color change of the suspensions from black to blackish brown obviously and at last to brownish yellow with increasing degree of oxidation.

3.1.1. XRD analysis

XRD observations can be used to characterize the crystal structure of the as-prepared GOs with varying oxidation levels [48]. According to Fig. 2(a), a diffraction peak at $2\theta = 26.28°$ can be observed for EG, indicating the distance between graphene layers is about 0.34 nm [49]. It is obvious that the intensity of diffraction peak at $2\theta = 26.28°$ decreases gradually with increasing oxidation degree of GOs, and eventually disappears at a high degree of oxidation. At the same time, a new diffraction peak appears at a smaller

Table 1
The concentration of different C groups in GOs.

Samples	C—C %At Conc.	C—OH %At Conc.	C—OO %At Conc.	C—O—C %At Conc.
EG	100	0	0	0
GO0.5	63.80	13.30	5.13	17.78
GO1	49.12	15.81	7.55	27.06
GO2	38.08	12.71	9.52	39.69
GO3	32.75	12.92	11.27	43.03
GO6	28.08	13.75	12.19	45.97

Table 2
The C/O ratio of EG and GO samples.

Samples	C1s %At Conc.	O1s %At Conc.	C: O
EG	95.03	4.97	19.12
GO0.5	83.82	16.18	5.18
GO1	78.11	21.89	3.57
GO2	73.58	26.42	2.79
GO3	70.74	29.26	2.42
GO6	37.04	62.96	0.59

angle. The XRD pattern of GO0.5 displays one slightly broadened diffraction peak around 2θ = 26.40° which indicates that the graphite lattice is distorted by mild oxidation [50]. The XRD pattern of GO1 displays two diffraction peaks at 2θ = 11.82° and 26.28°, showing that GO1 has the heterogeneous nature consisting of both sp^2 domains of graphite and sp^3 domains of graphite oxide [51]. The XRD pattern of GO2 shows that a diffraction peak shifts to 2θ = 10.38° and the diffraction peak from graphite disappears, indicating that there is no graphite in GO2 and the interlayer spacing of GO2 is about 0.85 nm. With further increasing the degree of oxidation, the XRD patterns of GO3 and GO6 show only the diffraction peak of GO at 2θ = 9.96° and 9.44° corresponding to around 0.89 and 0.93 nm of interlayer spacing, respectively. To sum up, graphitic domains remain in the samples with lower degree of oxidation and disappear in the samples with higher degree of oxidation while the interlayer spacing of the samples increases gradually with the increase in the oxidation levels.

3.1.2. XPS analysis

XPS can not only measure the element composition of sample surface, but also provide the chemical information of various elements. The XPS spectroscopy of GOs with respect to the degree of oxidation was measured for examining the different functional groups. Fig. 3 shows the C1s spectra of EG and GOs. There is only one peak corresponding to C—C stretching as a result of the sp^2 carbon bond in graphite at 284.6 eV in C1s spectra of EG, indicating there is no oxygenated functional groups [52]. The raw C1s spectra of GOs can be divided into 4 peaks corresponding to C—C, C—OH, C—O—C, C—OO, respectively, compared with the C1s spectra of EG. With the increase of the degree of oxidation, the intensity of C—C peak decreases gradually and the intensity of C—OH peak

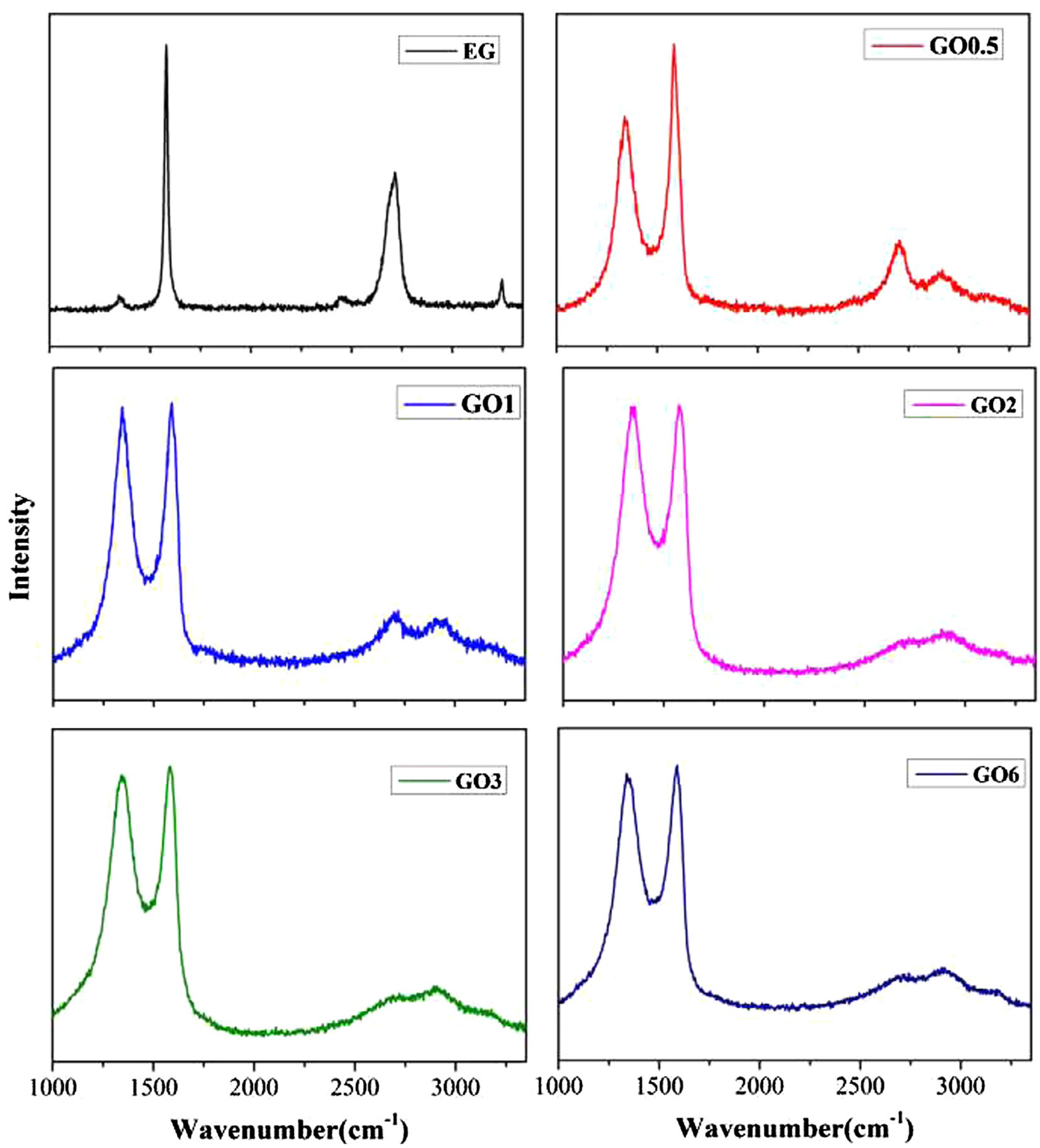

Fig. 4. Raman spectroscopy of GO with different degrees of oxidation.

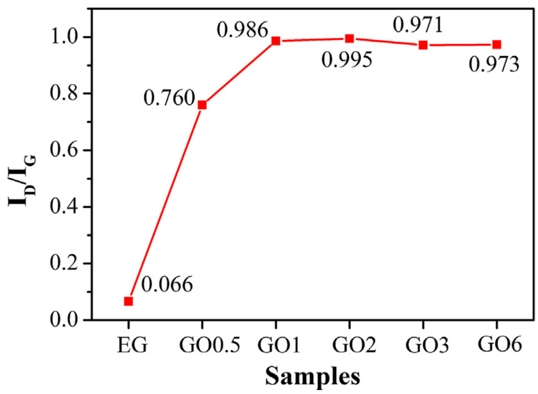

Fig. 5. Variation of I(D)/I(G) ratio with respect to the oxidation level of GOs.

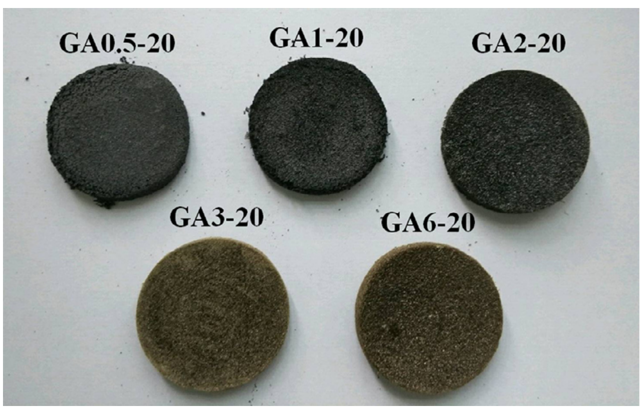

Fig. 6. Photographic images of GAs with different degrees of oxidation.

decreases slightly, but the intensities of C—OO and C—O—C increase.

To examine the functional groups of GOs with different degrees of oxidation, FTIR spectroscopy was performed, which can help to determine the types and structures of some or all of the molecules [53]. The transformation of oxygenated functional groups after oxidation can be seen from FTIR spectra in Fig. 2(b). With regard to EG, C—C bending in graphitic domains is seen at 1628 cm^{-1}. As seen from FTIR spectra, the peaks of C—OH (1403 cm^{-1}), C—O (~1050 cm^{-1}), C=O (1712 cm^{-1}), and C—O—C (1225 cm^{-1}) appear in turn with the oxidation levels of GOs increasing. When the mass ratio of EG: $KMnO_4$ is over 2, all the oxidized functional groups of GOs appear [54]. The concentration of different C groups of GOs can be calculated by casaXPS software from XPS spectroscopy, as listed in Table 1. Obviously, the concentrations of C—OO and C—O—C increase with the increment of oxidation degree, while that of C—C group decreases greatly. For GO1 and GO2, the concentration of C—OH is from 15.81% to 12.71% with the oxidation levels of GOs increasing, which implies that hydroxyl groups can be further oxidized. The results of FTIR and XPS indicates that some C—C groups are oxidized into C—OH, C—OO and C—O—C groups and hydroxyl groups can be translated into C—OO and C—O—C by further oxidation. The C/O ratios of EG and GO samples are listed in Table 2. From EG to GO6, the C atomic concentration decreases quickly with the rapid growing O atomic concentration. The C/O ratio decreases with the degree of oxidation increasing, which is in consistent with the previous results.

3.1.3. Raman analysis

Raman spectroscopy can be used to characterize of carbon material as a standard nondestructive tool [55]. As depicted in Fig. 4, Raman spectrum of EG reveals the presence of G band at 1580 cm^{-1} which corresponds to the first level scattering in the E_{2g} vibrational mode of graphene and reflects the neat sp^2 structure [56]. In front of G band, there is a small peak around 1345 cm^{-1} named D band which shows the defects in the graphene sheet [51,57]. Apart from them, the spectrum contains another important feature of EG, a 2D band at 2700 cm^{-1}. The 2D band is more sensitive to characterize neat π-π conjugated structure in graphite [58], which also displays a striking change depending on the degree of oxidation. Also, compared with the Raman spectrum of EG, the intensity of D band of GOs is significantly stronger, which demonstrates that the structure of graphene was destroyed during oxidation. Furthermore, the intensity of the 2D band tends to decrease and broaden gradually with the increasing of oxidation levels. In Fig. 5, the GOs with lower oxidation levels show a clear increase in $I_{(D)}/I_{(G)}$ ratio. The $I_{(D)}/I_{(G)}$ ratio becomes saturated at higher oxidation degrees of GOs. These results illustrate that the disruption of graphitic stacking order in GO is more and more serious with increasing oxidation level [59,60].

3.2. The morphologies of 3D GAs

To get the ultra-light porous skeletons of GO, GO solutions were lyophilized to fabricate GAs. However, the degree of oxidation of GO0.5 and GO1 was too low to form a complete framework of GAs, as shown in Fig. 6. The diameters of GA0.5 and GA1 are smaller than that of the cylindrical mold. Both samples are very fragile as seen from the powders dropped from the edge. Hence, the other three GAs from GOs of different degrees of oxidation are examined hereafter.

Fig. 7 shows the SEM images of GA2, GA3 and GA6. The GAs exhibit an interconnected porous 3D framework of crinkly sheets with many macropores owing to GO sheets cross-linked by hydrogen bonding interactions and water freezing. A specific crinkly structure of GO can be seen in the SEM images of as-synthesized GAs. The formation of 3D crinkly shape is from sp^3 carbon atoms, and results in the increase of interlayer spacing of GO. But in Fig. 7a, it can be seen that the sheets of GA2 are still stacked partly. In Fig. 7c, better porous 3D framework of carbon can be observed and the structure of the rib is relatively intact. With further oxidation, there are a certain breakages in the ribs as shown by the red circles in Fig. 7e.

3.3. The properties of PEG and composite PCMs

3.3.1. Shape stability of PEG and the composite PCMs

The shape stability of the composite PCMs is characterized by leakage test first. Pure PEG and composite PCMs are heated from 30 to 70 °C on a hot stage directly. As shown in Fig. 8, all the samples maintain their original defined shapes at 30 °C, below the melting point (65 °C) of PEG. When heated to 70 °C for about 3 min, pure PEG melts gradually and flows, while PGA2-40 melts and infiltrates the filter paper slightly, and PGA3-40 starts to melt but the leakage even cannot wet the filter paper, and PGA6-40 show none indication of leakage. After 5 min at 70 °C, pure PEG has been molten completely, however, PEG/GA composites melts with little liquid leakage at the bottom owing to the confinement effect of GAs.

Cylindrical composite PCMs with the dimension of 25 mm in diameter and 2.5 mm in thickness are prepared and loaded in a dynamic rheometer with parallel-plate geometry under a constant force (7 N) with the temperature rising from 35 °C to 150 °C. In the heating process, the dimension (thickness) change of the PCMs during melting is recorded by the dynamic rheometer. As seen from Fig. 9, all the composite PCMs show a slight increase on dimension at the beginning of the heating owing to the thermal

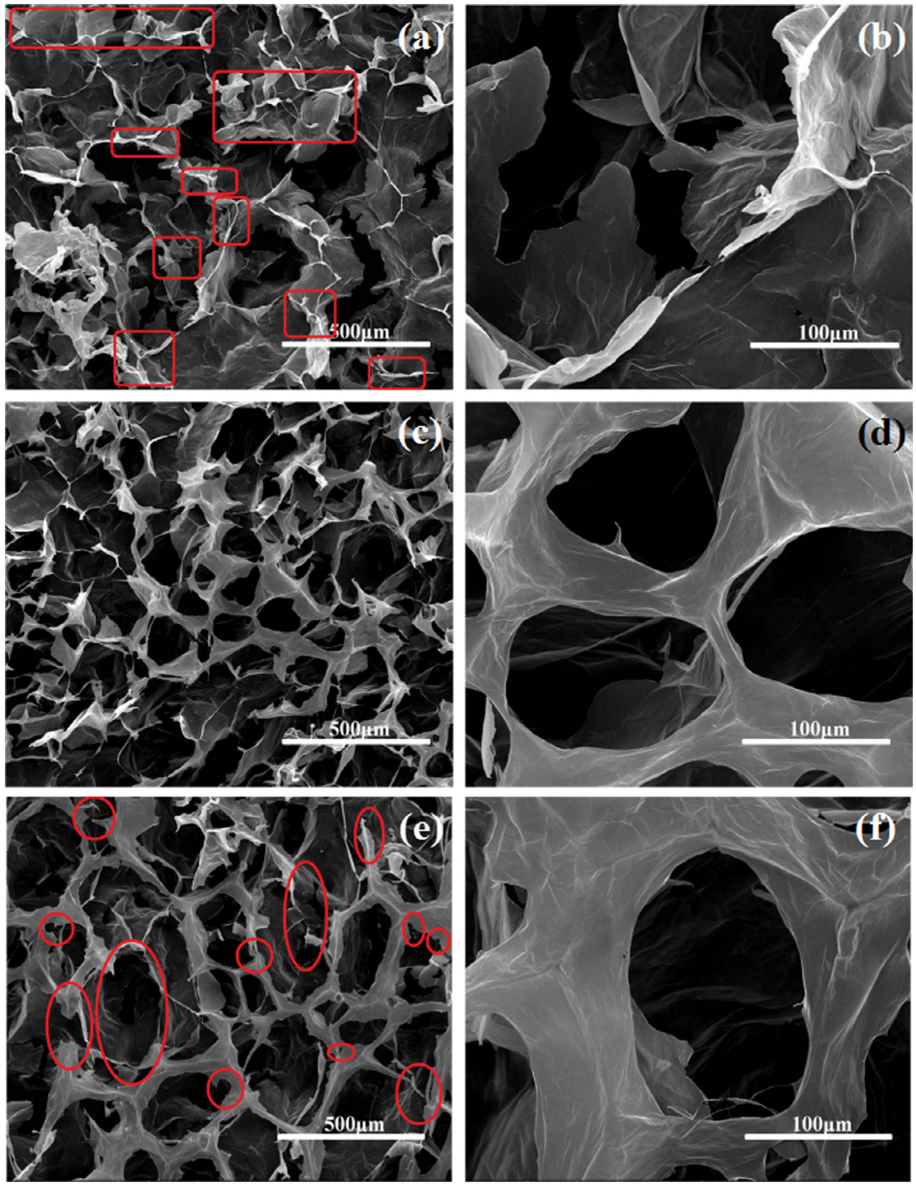

Fig. 7. SEM images of GA 2 (a) and (b), GA3 (c) and (d), GA6 (e) and (f). The initial concentration of the GOs aqueous solutions was 20 mg/mL.

expansion of the samples. When the temperature reaches 63 °C (around T_m of pure PEG), the dimension of pure PEG reduces sharply until close to zero, indicating that pure PEG has completely leaked. On the contrary, all the composite PCMs show a great shape stabilizing effect, whose dimensions change a little even at a temperature as high as 150 °C due to the existence of the inherent 3D network structure of GAs. Especially, when the rheological tests are carried under a 7 N force upon T_m of PCM, the composite PCMs still keep their shapes. The minimum dimensions of PGA2-40, PGA3-40 and PGA6-40 is 90.3%, 99.3% and 100.1% of their original dimensions, respectively. The composite PCMs show better shape stabilizing effect with increasing oxidation level of GAs. The confinement effect of composite PCM mainly comes from two aspects. One is the existence of the 3D network structure and the other is the interaction of hydrogen bonding between working medium and supporting material [22,61].

The hydrogen bonding of PEG/GA composite PCMs was characterized by FTIR spectroscopy (Fig. 10). Seen from Fig. 10, C—O peaks (around 1110 cm^{-1}) in all the composite PCMs shift to lower wavenumbers with oxidation degrees of GAs increasing (from PEG to PGA210, PGA310, PGA640, from 1107.1 cm^{-1} to 1106.5, 1106.0, 1105.8 cm^{-1}, correspondingly), which indicates the stronger interactions of intermolecular hydrogen bonding between PEG and GOs. Similar phenomena have also been reported in literatures [22,62,63]. It is proved that hydrogen bonding is a reason for GAs to impede the leakage of PEG melt.

3.3.2. Thermo-physical properties of PEG and the composite PCMs

Shape stability of the PEG is one of basic requirements as phase change energy storage materials, which can be the significantly improved by introducing GA. Another basic requirement that cannot be ignored is the high energy storage density, which is decreased for sure when the content of fillers in the composite PCMs increases. Fig. 11 shows the DSC curves of all the samples with different degrees of oxidation of GOs, and the peaks on the DSC curves correspond to the solid–liquid phase transformation.

The area under the endothermal/exothermal peaks on the curves represents the phase change enthalpy of the shape-stabilized PCMs. The detailed results of phase transition characteristics obtained for PEG and the three composite PCMs including

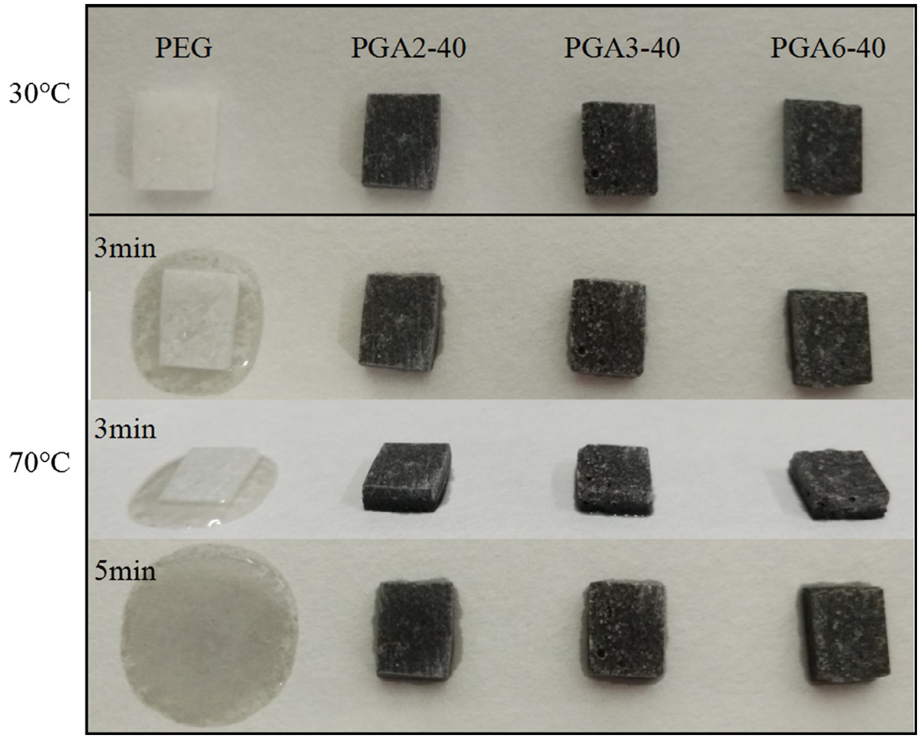

Fig. 8. Shape-stability of PEG and PEG/GAs composite PCMs with increasing temperature.

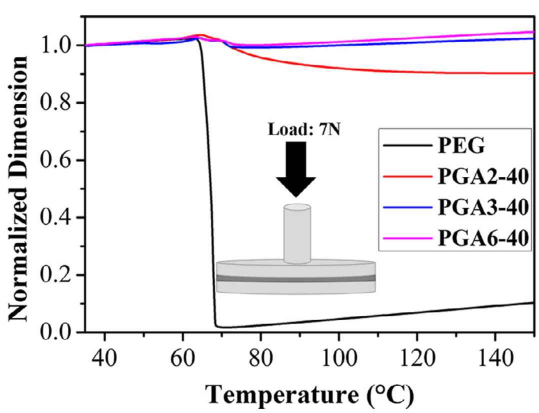

Fig. 9. Shape stabilizing effect of pure PEG and composite PCMs. The curves of dimension change vs. temperature were obtained from a temperature scanning under constant load (7 N) in a dynamic rheometer.

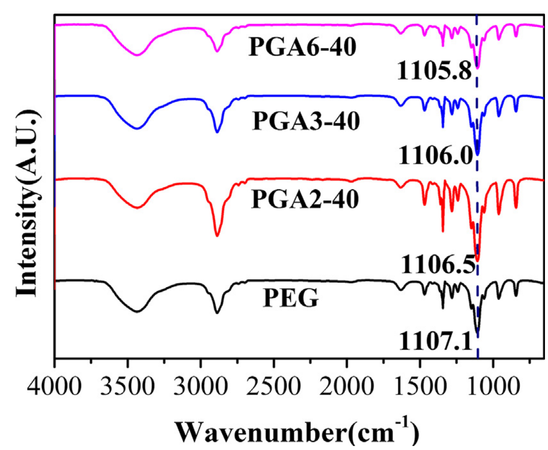

Fig. 10. FTIR patterns of PGAs with different oxidation degrees of GAs.

onset/end crystallization temperature (T_{co}/T_{ce}), onset/end melting temperature (T_{mo}/T_{me}), peak melting/crystallization temperature (T_{mp}/T_{cp}) and melting/crystallization enthalpy ($\Delta H_m/\Delta H_c$) are listed in Table 3. Compared with pure PEG, there is a decrease of T_{cp} and an increase of T_{mp} of the composite PCMs which results from the confinement effect of GAs [64,65] and the phase change enthalpy of composite PCMs are slightly reduced owing to the extremely low filler loading which replaces minimal working materials [66] and the fact that the confined movement of the molten PEG chains from GAs hindered the crystallization and melting of PCM [61].

3.3.3. Thermal reliability of composite PCMs

The thermal stability of the composite PCMs was further characterized by 500 DSC heating and cooling cycles. After 0, 20, 50, 100, 200, 500 heating and cooling cycles, the DSC curves of PEG/GA composite PCMs are presented in Fig. 13(a). The change of endothermal and exothermal curves is tiny after 500 time thermal cycles. The detailed results of T_{mo}/T_{co}, T_{me}/T_{ce}, T_{mp}/T_{cp}, $\Delta H_m/\Delta H_c$ for PGA3-20 after 0, 20, 50, 100, 200, 500 heating and cooling cycles are given in Table 4. After 20, 50, 100, 200 and 500 thermal cycles, all the T_{mp} and T_{cp} of PGA3-20 are similar and change by ac. 4 °C and 1 °C compared with original PGA3-20, respectively. This is because a little PEG is adsorbed onto the surface of samples and the composite PCMs show some leaks on the surface, which can be clearly observed in the SEM images of composite PCMs before and after heating (Fig. 12). The surface of PGA3-20 before heating is flat, but the surface of PGA3-20 after heating is uneven and shows a lot of depressions which indicates that a part of PEG adsorbed onto the surface of samples melts and leaks upon its melting temperature. However, the cross-section of PGA3-20 after heating is similar with that of PGA3-20 before heating, which implies that the composite PCM has a better 3D network to prevent the leakage of molten PEG.

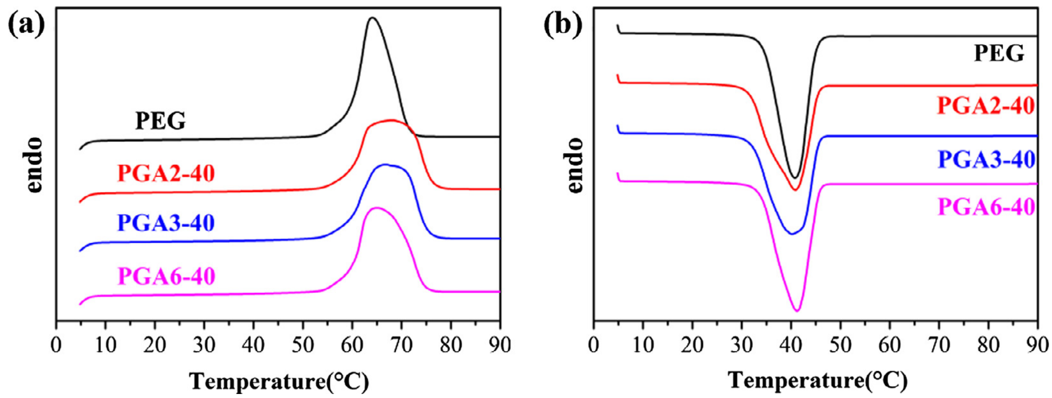

Fig. 11. DSC heating scan (a) and cooling scan (b) of pure PEG and PEG/GA composite PCMs.

Table 3
DSC heating and cooling characteristics of PEG and PGA composite PCMs.

Samples	T_{co} (°C)	T_{ce} (°C)	T_{cp} (°C)	ΔH_c (J/g)	T_{mo} (°C)	T_{me} (°C)	T_{mp} (°C)	ΔH_m (J/g)
PEG	45.01	40.80	34.45	174.6	59.38	64.15	71.40	182.9
PGA2-40	45.51	40.88	31.52	169.2	59.21	67.96	76.01	176.0
PGA3-40	45.57	40.20	32.09	167.0	58.67	66.76	75.08	173.7
PGA6-40	45.59	41.20	33.78	170.1	59.57	65.06	74.53	177.7

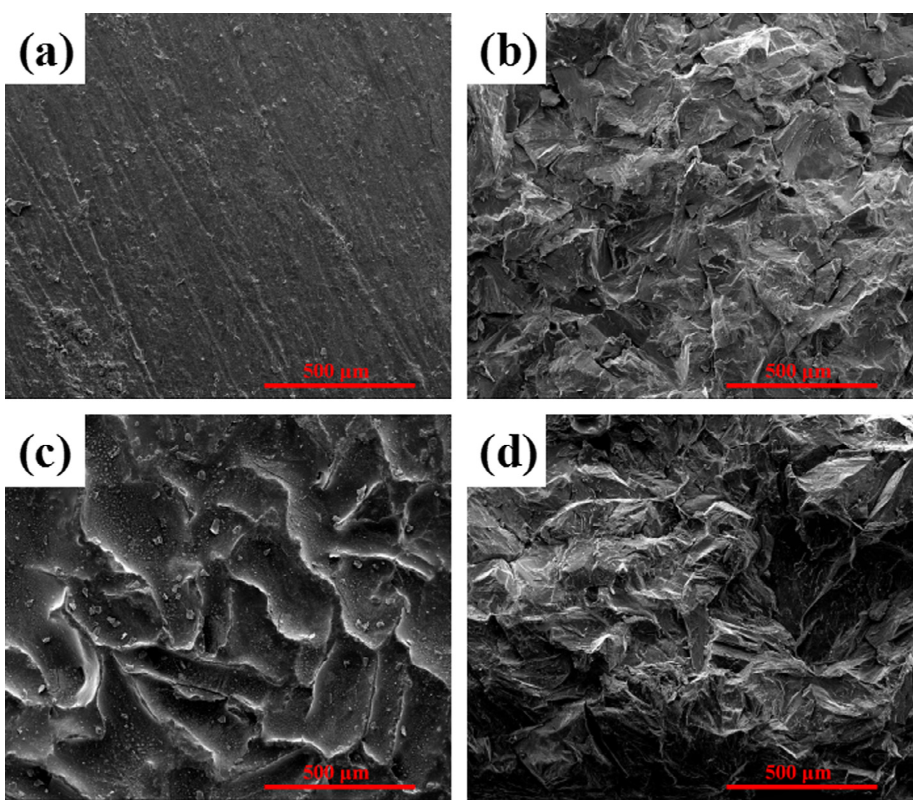

Fig. 12. SEM images of the surface (a) and cross-section (b) of PGA3-20 before heating, and the surface (c) and cross-section (d) of PGA3-20 after heating.

The melting enthalpy changes by 0.25%, 0.80%, 2.35%, 4.06% and 6.31% and the crystallization enthalpy changes by 0.42%, 0.99%, 2.36%, 4.08% and 6.49% corresponding to the 20, 50, 100, 200 and 500 heating and cooling cycles, respectively, which is acceptable for LHTES applications. Additionally, Fig. 13(b) shows the FTIR spectrum of PGA3-20 after cycling of 500 times. No appearance or disappearance of new peak occurs after 500 heating and cooling cycles, revealing that the chemical structure of the composite PCM is also stable. Based on the above results, it can be concluded that the composite PCMs possess commendable thermal reliability.

3.3.4. Photo-to-thermal energy conversion

PEG and composite PCMs were exposed to 100 mW cm^{-2} light irradiation for 20 min and then the light was removed. Fig. 14(a)

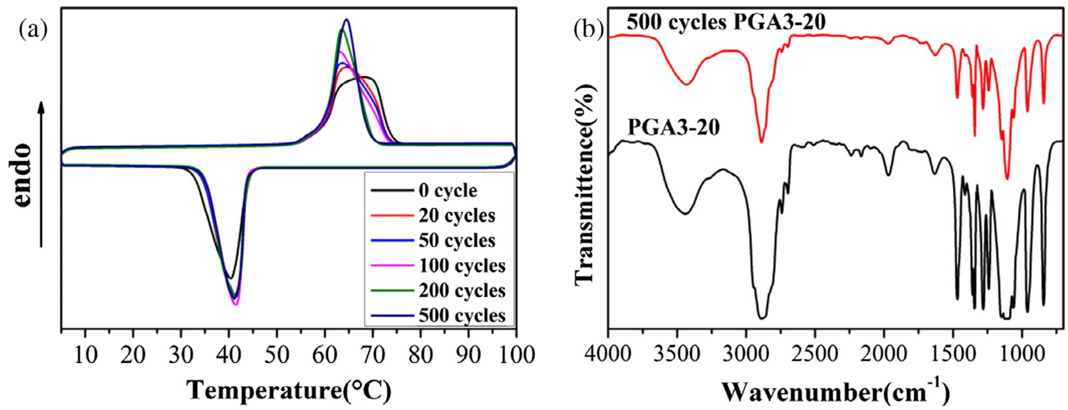

Fig. 13. (a) The DSC heating and cooling curves of PGA3-20 after 0, 20, 50, 100, 200 and 500 cycles. (b) FTIR spectrum of PGA3-20 with cycling of 500 times.

Table 4
DSC heating and cooling characteristics of PGA3-20 after 0, 20, 50, 100, 200 and 500 cycles.

Cycle Numbers	T_{co} (°C)	T_{ce} (°C)	T_{cp} (°C)	ΔH_c (J/g)	T_{mo} (°C)	T_{me} (°C)	T_{mp} (°C)	ΔH_m (J/g)
0	43.94	32.46	40.34	191.0	59.28	73.83	68.06	199.6
20	43.65	34.52	41.19	190.2	59.80	73.32	64.60	199.1
50	43.70	34.22	41.15	189.1	59.95	73.35	63.70	198.0
100	43.64	34.97	41.33	186.5	60.03	73.39	63.22	194.9
200	43.54	35.19	41.37	183.2	60.06	70.00	63.53	191.5
500	43.45	35.10	41.11	178.6	60.17	68.76	64.50	187.0

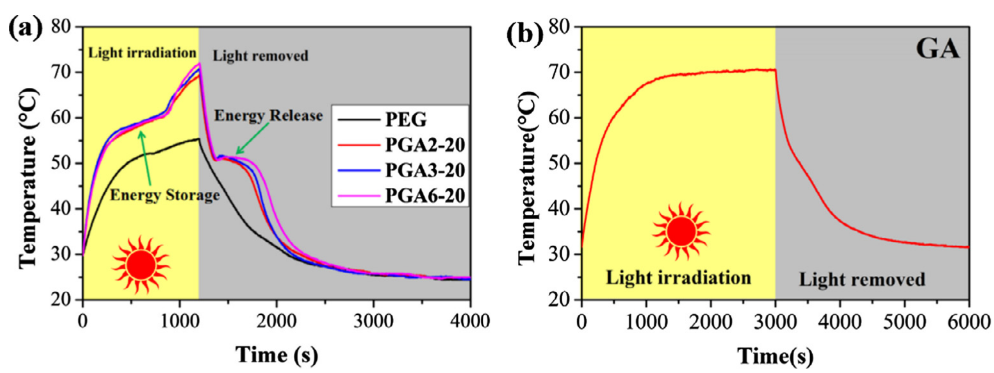

Fig. 14. Temperature evolution curves of (a) pure PEG, PEG/GA composite PCMs and (b) GA under light irradiation.

shows temperature evolution curves of all the samples. As observed, the temperature of PEG/GA composite PCMs explodes under light irradiation and the temperature rising speed of composite PCMs is significantly higher than that of pure PEG, which can be attributed to the better light absorbing ability of PEG/GA composite PCMs from their dark appearance. The white surface of pure PEG reflects the irradiating light, leading to poor photoabsorption. The photo-to-thermal conversion effect of PCMs can be enhanced by introducing dark materials [44]. Here the photon capture and photo-to-thermal energy conversion have been realized by GAs as photon antennas [43]. A temperature inflection point appears for PEG/GA composite PCMs at the phase change temperature of PEG, which indicates that thermal energy storage in the process of phase change takes place with a slight increase of temperature. Thus, the photo-to-thermal energy storage has been realized. After illumination for 20 min, the temperature of PEG/GA composite PCMs reaches around 70.0 °C which is higher than the melting point of PEG, while the temperature of pure PEG is only 55.4 °C, below its phase change temperature. After turning off the light, the temperatures of composite PCMs reduce suddenly and speedily. Subsequently, another platform appears when the temperature is around the crystallization point, indicating the release of a large amount of stored thermal energy. After the completion of phase change process, the temperatures of the PCMs continue to decrease rapidly, and then the pace of decline gradually slows down. On the contrary, the curves of pure PEG and GA bulk show no plateau, because pure PEG has too poor photoabsorption to reach the phase change temperature and there is no phase change process for GA (Fig. 14b). Compared with pure PEG, the composite PCMs show a more excellent property in photo-to-thermal energy conversion and storage for potential applications.

4. Conclusion

GOs with various oxygenated functional groups and chemical structures are synthesized by tuning different oxidation degrees using a modified Hummers' method. Different oxygen functional groups are formed in the various stages of oxidation and the effect of them on the structural and chemical analysis is researched. The graphitic nature of the GOs decreases but species of oxygenated functional groups increases with the increase of oxidation level. The GAs prepared by freeze-drying GO suspension exhibits an interconnected porous 3D framework of crinkly sheets with many macropores. The PEG/GA composite PCMs with superior shape stability, high energy storage density, desirable thermal reliability, and more efficient photo-to-thermal energy conversion has been fabricated by incorporation of GAs with various structures. Under the load of a constant force (7 N), the dimension of PGA2-40 heated from 35 °C to 150 °C is 90.3% of their original dimension. Interestingly, the change of dimension for PGA3-40 and PGA6-40 is negligible. These results demonstrate that the higher the oxidation level is, the better the shape-stability is, after that a balance is reached. The reason for the difference is most likely capillary force of the porosity and hydrogen bond between PEG and oxygenated functional groups of GO increasing with the increase of the oxidation degree.

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